



**Project
Management
Institute.**



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PROJECT MANAGEMENT

IREC supports your career journey.

Webster dictionary definition.

- Definition of *agile*:

- **1:** marked by ready ability to move with quick easy grace an *agile dancer*.
- **2:** having a quick resourceful and adaptable character an *agile mind*.



Adapting project management approaches to dynamic environments - AGILITY ■

Outline:

- I. The project environment.
- II. Predictable and Dynamic project environments.
- III. Examples of uncertainty and variability.
- IV. Dynamic environment – VUCA.
- V. Managing projects using an adaptive approach.
- VI. The Adaptive/Agile mindset.
- VII. Agile Characteristics – Ex. Methodologies.

The Project Environment ■

The project environment is multifaceted & includes the:

- *organizational setting,*
- *stakeholder influences,*
- *regulatory and legal requirements,*
- *technological resources,*
- *economic and market conditions,*
- *social and cultural factors,*
- *project constraints,*
- *and environmental considerations.*

*****Understanding these elements helps project managers effectively plan, execute, and manage projects.**

Predictive approach - Waterfall - Traditional.

- I. Predictive Approach:** Emphasizes detailed upfront planning, strict adherence to initial plans, and sequential execution etc.....
- II. Changes** are managed through formal processes and are generally minimized.

A predictable project environment - Ideal.....



Uncertainty & Variability - the reality....

A real project environment is dynamic – marked by frequent changes and uncertainties in various aspects such as requirements, technology, market conditions, stakeholder interests, regulations, project complexity, team dynamics, organizational shifts and external factors (political-economic-social).

Uncertainty & Variability - Poor visibility.



Uncertainty and Variability examples ...

1. Requirement Changes.

- **Example:** A software development project starts with a clear set of requirements, but midway through, the client decides to add new features or change existing ones based on new market trends or user feedback.
- **Impact:** *This can lead to scope changes, increased costs, and delays in the project timeline.*

Uncertainty and Variability examples ...

2. Technological Advancements.

- **Example:** A construction project is planned with certain technologies in mind, but during the project's lifecycle, a new technology emerges that could significantly enhance efficiency or quality.
- **Impact:** *The project may need to adapt to new technology, which could involve additional costs, rework, or delays.*

Uncertainty and Variability examples ...

3. Resource Availability.

- **Example:** A project team relies on external contractors whose availability fluctuates due to their other commitments or unexpected issues.
- **Impact:** *Variability in resource availability can lead to delays and difficulties in meeting project milestones.*

Uncertainty and Variability examples ...

4. Construction Project Timelines.

- **Case:** A residential building is being constructed.
- **Variability:** The duration of various construction tasks can vary due to weather conditions, the availability of subcontractors, and the efficiency of the construction crew. These fluctuations can lead to deviations from the original project timeline.

Uncertainty and Variability examples ...

5. New Product Development (NPD).

- **Case:** A tech company is developing a groundbreaking wearable device.
- **Uncertainty:** The project faces uncertainty regarding how consumers will respond to the new product, as it's a novel concept with no existing market precedent. Additionally, there is uncertainty about whether the technology will work as expected and whether it will comply with regulatory standards.

We live in a dynamic environment - VUCA.

Volatile
Uncertain
Complex
Ambiguous

"the new
normal"

VUCA Environment.

VUCA

Volatile

The environment demands you react quickly to ongoing changes that are unpredictable and out of your control

Uncertain

The environment requires you to take action without certainty

Complex

The environment is dynamic, with many interdependencies

Ambiguous

The environment is unfamiliar, outside of your expertise

Use the concept of ‘Tailoring’.

- *Tailoring is the deliberate adaptation of the project management approach, governance, and processes to make them more suitable for the given environment and the work at hand.*
- **Each project is unique.** Adapt methods to the unique project context to determine the most appropriate way of working to produce the desired outcomes
- Tailor iteratively and continuously throughout the project

Managing projects using an Adaptive approach.

1. Predictive, Plan-driven, Water-fall, Traditional.

- A form of project life cycle in which the project scope, time, and cost are determined in the initial phases of the life cycle

2. Agile, Adaptive, Change-driven.

- The Agile methodology is a way to manage a project by breaking it up into several phases. It involves constant collaboration with stakeholders and continuous improvement at every stage using short feedback loops.

3. Hybrid. A combined approach that uses a strategy from agile or predictive for a specific need. Project might switch approaches based on need, changing work requirements, or circumstances.

Principles Behind the Agile Manifesto

1 to 6

1. Our highest priority is to **satisfy the customer through early and continuous delivery** of valuable software.
2. **Welcome changing requirements**, even late in development. Agile processes harness change for the customer's competitive advantage.
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a **preference to the shorter timescale**.
4. **Businesspeople and developers must work together** daily throughout the project.
5. Build projects around **motivated individuals**. Give them the environment and support they need and trust them to get the job done.
6. The most efficient and effective method of conveying information to and within a development team is **face-to-face conversation**.

Principles Behind the Agile Manifesto

7 to 12

7. Working solution is the primary measure of progress.
8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
9. Continuous attention to technical excellence and **good design enhances agility.**
10. Simplicity – the art of maximizing the amount of work not done – is essential.
11. The best architectures, requirements, and designs emerge from **self-organizing teams.**
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

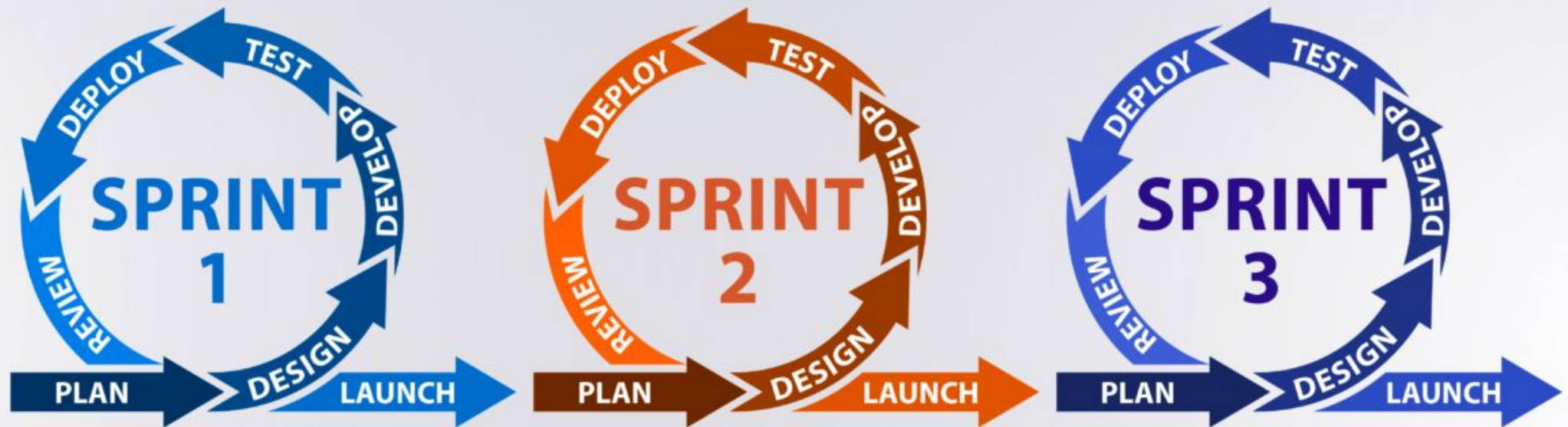
Managing projects using an Adaptive approach.

- **Adaptive/Agile Approach:** Emphasizes flexibility, iterative development, and responsiveness to change.
- **The project plan evolves based on feedback and changing conditions, allowing for frequent adjustments and continuous improvement.**

Managing projects using an Adaptive approach.

1. **Flexible Scope and Plan** : The project scope and plan are developed at a high level initially and refined iteratively based on feedback and changing conditions.
2. **Iterative Phases** : The project is divided into iterative cycles or sprints, where each cycle delivers incremental progress and allows for reassessment and adjustments.
3. **Change Embrace** : Changes are expected and welcomed. The project adapts to new information and feedback, making ongoing adjustments to align with evolving needs.

Work done in short cycles (sprints) to allow feedback and adaptation.



Managing projects using an Adaptive approach.

4. Minimal Documentation : Documentation is kept minimal and evolves as needed. The focus is on delivering working solutions and incorporating feedback rather than extensive documentation.

5. Enhancing Team Collaboration and Morale:

- ***Empowerment***: Adaptive methodologies often empower team members to make decisions and contribute ideas. This increased autonomy can enhance job satisfaction and motivation.
- ***Collaborative Environment***: Regular meetings, such as daily stand-ups in Agile, foster communication and collaboration, helping teams work together more effectively.

Managing projects using an Adaptive approach.

6. Facilitating Innovation.

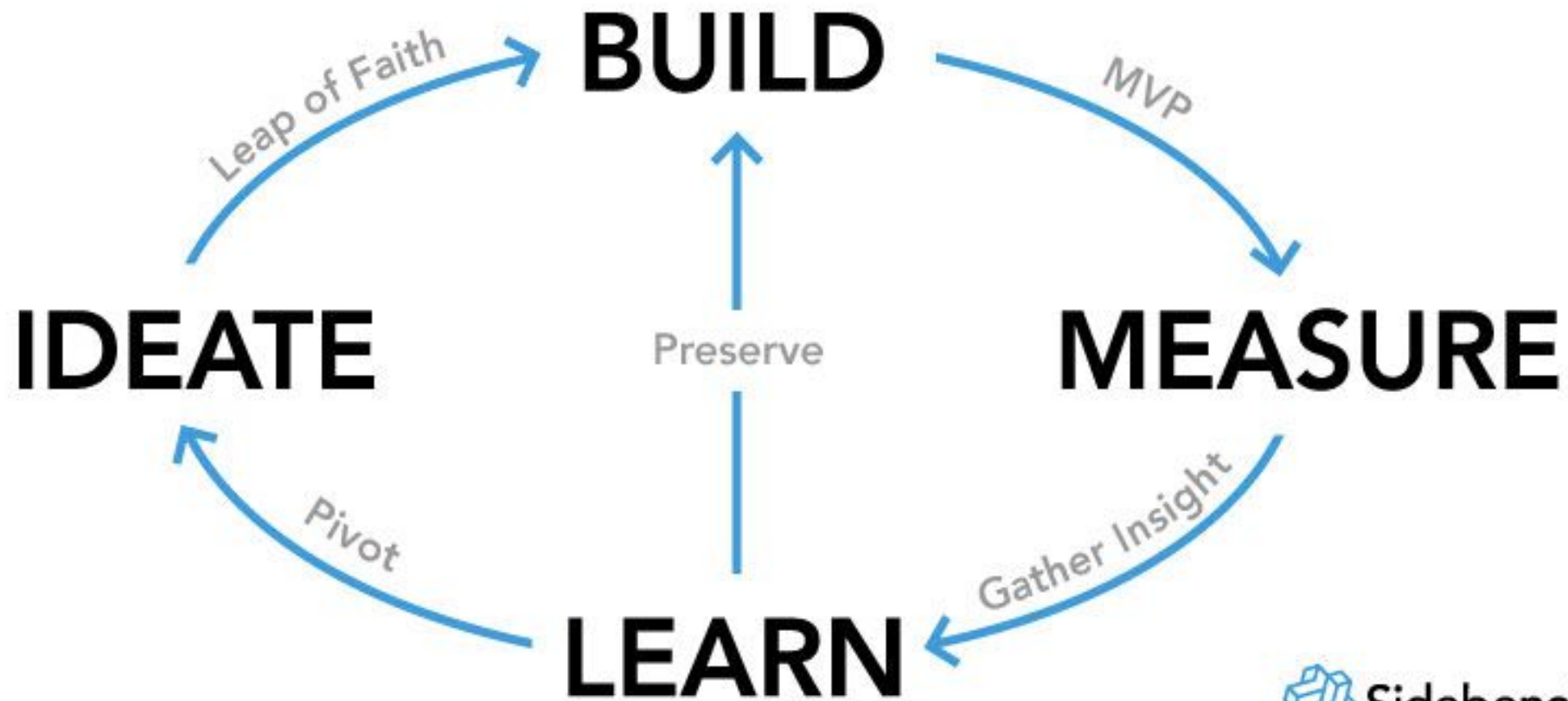
- **Iterative Development:** Adaptive approaches encourage experimentation and innovation by allowing for iterative development and testing of new ideas.
- **Feedback-Driven Changes:** Teams can quickly pivot or refine innovations based on feedback, fostering a culture of continuous improvement and creativity.

7. Improving Risk Management.

- **Early Detection:** Adaptive project management allows for the early identification and mitigation of risks. Regular reviews and iterations help in spotting potential issues before they become major problems.
- **Mitigation Strategies:** With frequent checkpoints and feedback loops, teams can implement mitigation strategies promptly, reducing the impact of risks.

Agile approach thrives on “Fail Fast” Approach.

❑ *Prototyping, Mock-ups, MVPs, Experimenting etc.*



Incremental delivery of useable product features.

- ❑ Minimum Viable Products – *early return on investment.*



Incremental delivery of useable product features.

- ❑ Minimum Viable Products – *early return on investment.*

Software Development

- **Example:** A mobile app that starts with a basic login feature and gradually adds functionalities like user profiles, chat, and notifications in successive updates.
- **Benefit:** *Users can start interacting with the app early, providing feedback that helps refine the next features.*

Incremental delivery of useable product features.

- ❑ Minimum Viable Products – *early return on investment.*

Automotive Industry

- **Example:** A car manufacturer releases a basic electric vehicle with core driving features. Over time, they add software updates for features like autonomous driving or enhanced infotainment.
- **Benefit:** *Customers receive the basic vehicle early, and the product improves through over-the-air software updates.*

Incremental delivery of useable product features.

- ❑ Minimum Viable Products – *early return on investment.*

Healthcare Technology

- **Example:** A healthcare software platform that first releases a patient appointment scheduling system. Later, it adds functionalities like electronic health records (EHR), telehealth, and billing modules.
- **Benefit:** *Healthcare providers can use the essential scheduling tool immediately while awaiting additional features.*

Agile work spaces encourage collaboration.



Agile uses Daily Stand-ups to track progress.



Agile uses Daily Stand-ups to track progress.

Enhanced Accountability

- **Example:** During the meeting, each team member discusses what they completed yesterday and what they plan to achieve today.
- **Benefit:** *This creates a sense of accountability, as team members publicly commit to their tasks, encouraging follow-through.*

Agile uses Daily Stand-ups to track progress.

Quick Problem Identification

- **Example:** A developer mentions they're stuck on a technical issue during the stand-up. Another team member who has encountered a similar issue offers a solution right away.
- **Benefit:** *Daily stand-ups help identify blockers early and allow teams to address issues promptly, avoiding delays.*

Face-to-Face Communication

Solve
Problems
Rapidly



Knowledge
Sharing

Agile work spaces encourage collaboration.

Improved Communication

- Agile encourages open, frequent communication within the team, ensuring that everyone is aligned on project goals, progress, and obstacles.
- *Regular feedback loops and collaboration reduce misunderstandings, leading to smoother project execution and fewer delays.*

Agile work spaces encourage collaboration.

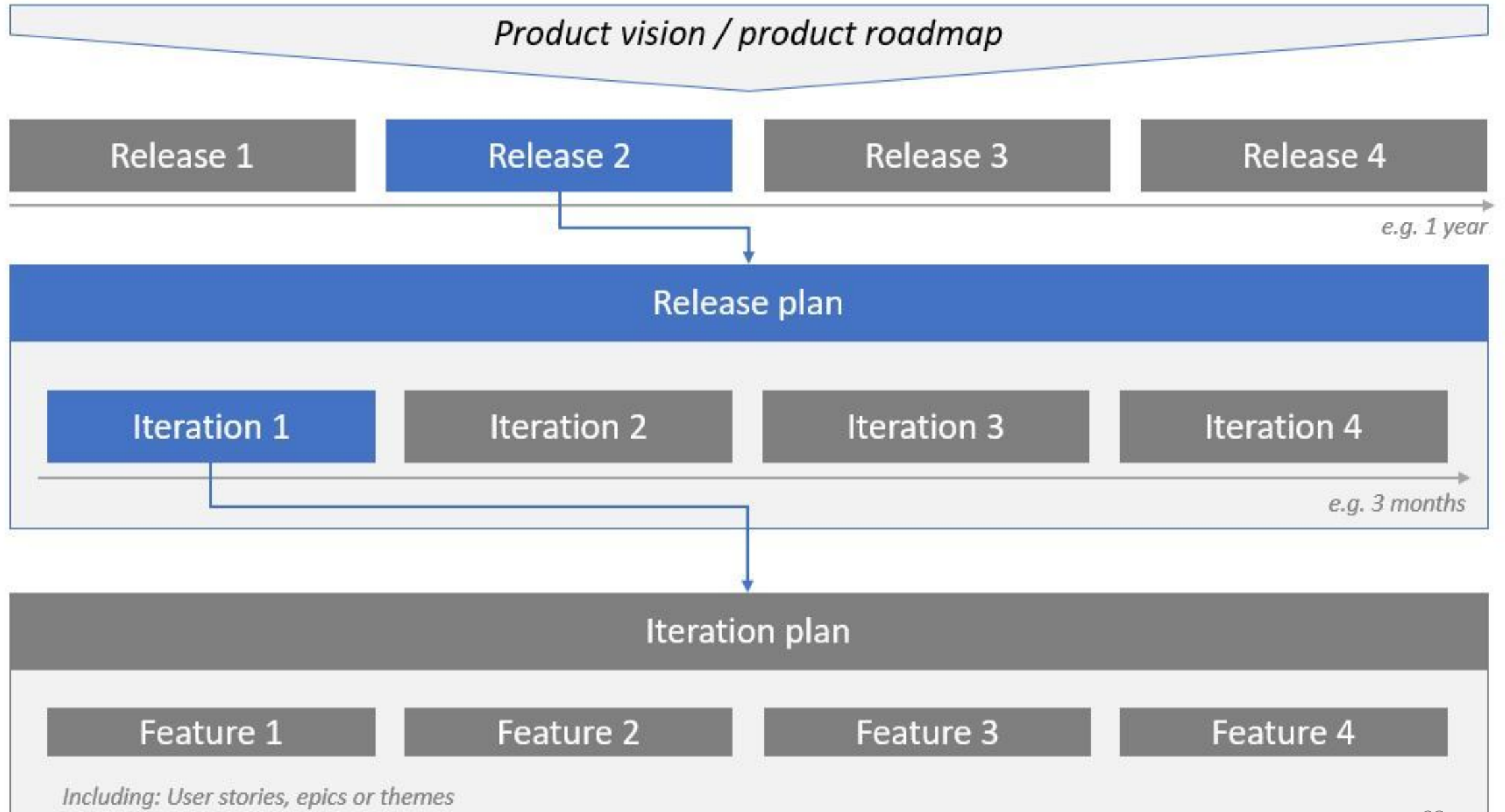
Enhanced Flexibility and Adaptability

- Collaborative teams are more adaptable to changes. Agile's iterative nature allows teams to pivot quickly based on customer feedback or shifting priorities.
- *Teams can self-organize and find the best ways to respond to changes, ensuring that the project stays on track.*

Agile work spaces encourage collaboration.

Shared Ownership and Accountability

- In collaborative Agile teams, ownership of tasks and goals is shared, which fosters a sense of collective responsibility.
- *Everyone in the team is accountable for delivering value, and this shared accountability drives higher commitment and motivation.*



Release 1



Sprint 1

Sprint 2

Sprint 3

Release 2



Sprint 4

Sprint 5

Sprint 6

Release 3



Sprint 7

Sprint 8

Sprint 9

Conclusion.

- ❑ **Scrum** : Roles + Meetings + Documents (artifacts).
- ❑ Kanban,
- ❑ Lean Agile,
- ❑ XP,
- ❑ FDD,
- ❑ Crystal etc.

AGILE TERMS.

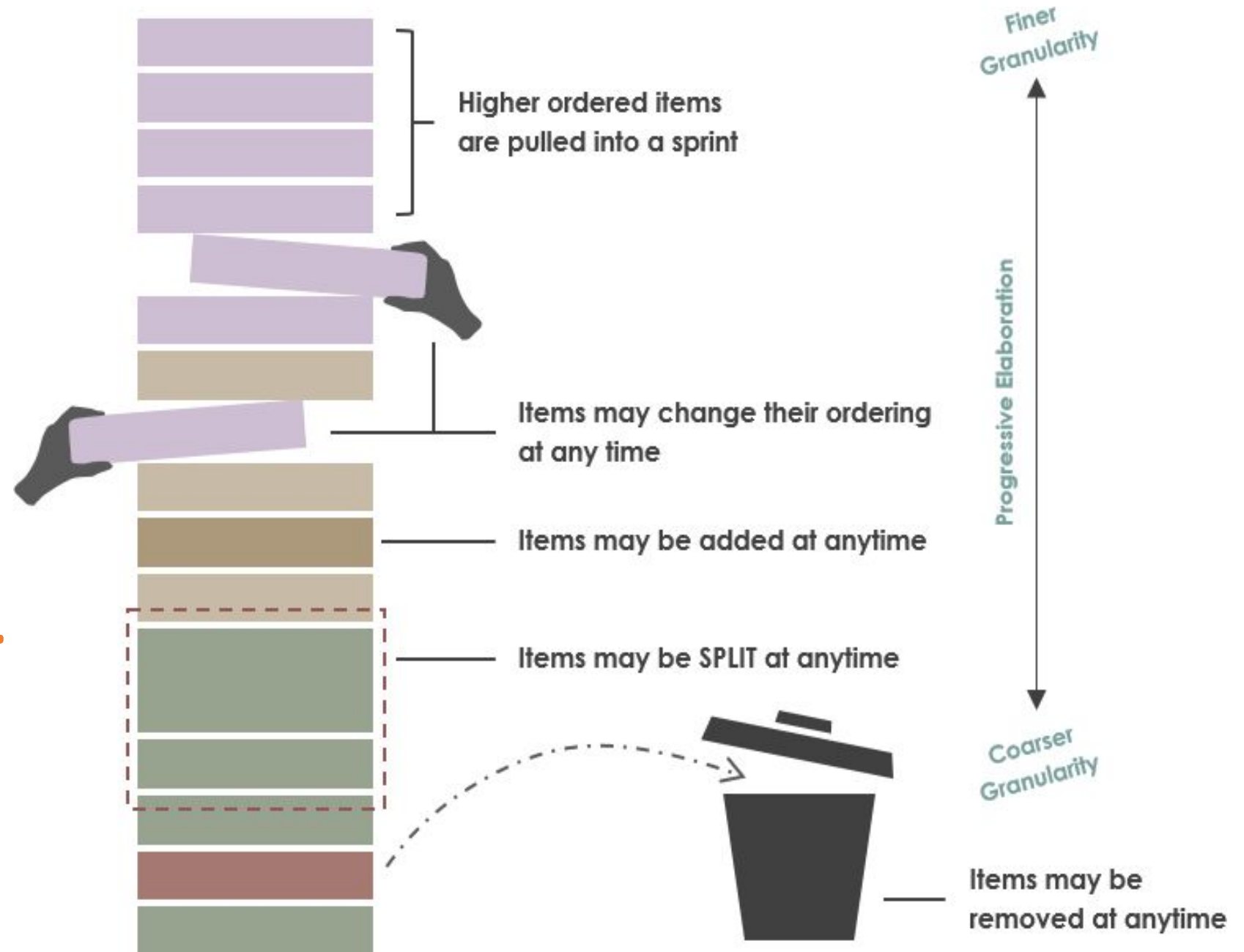
- **User stories:** A very high-level definition of a **requirement**, containing just enough information so that the developers can produce a reasonable estimate of the effort to implement it.
- **Product Backlog:** An ever-evolving **list of product requirements**, prioritized by the customer (or customer representative), that conveys which features to implement.
- **Iteration:** Uninterrupted '**period of time**' during which the team performs work at the end of which the team delivers a 'potentially shippable product'. Also called a sprint.

Writing a user story

- 1 Define your **end user**
- 2 Specify what **they want**
- 3 Describe **the benefit**
- 4 Add **acceptance criteria**

<https://blog.csdn.net/chikisang>

Product Backlog Prioritization.



Agile Terms.

- **An epic:** Is a large user story that cannot be delivered as defined within a single iteration or is large enough that it can be split into smaller user stories.
- **Time Boxing:** A time management technique that helps organize the performance of work and manage scope. A time box is a fixed-length of time during which an activity is done.
- **Sprint planning meeting:** Is an event in agile where the team determines/identifies the product backlog items, they will work on during that sprint and discusses their initial plan for completing those product backlog items.

Agile Terms.

- **Definition of Ready (DoR):** Involves creating clear criteria that a user story must meet before being accepted into an upcoming iteration. The Team determines how something is done and the amount of work required to do it.
- **Definition of Done (DoD):** It describe a list of criteria that must be completed for a project stage to be considered “done. It is a formal description of the state of the Increment when it meets the quality measures required for the product.

Agile terms.

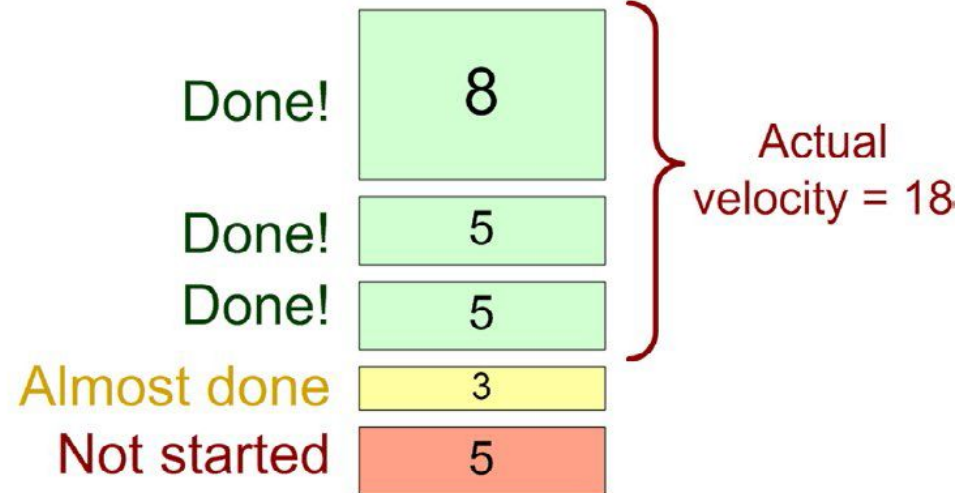
- **Velocity.** This is a measure of how much work is completed by the Developers within a specific Sprint. Accordingly, it is also a measure of the speed of a development team.

Beginning of sprint



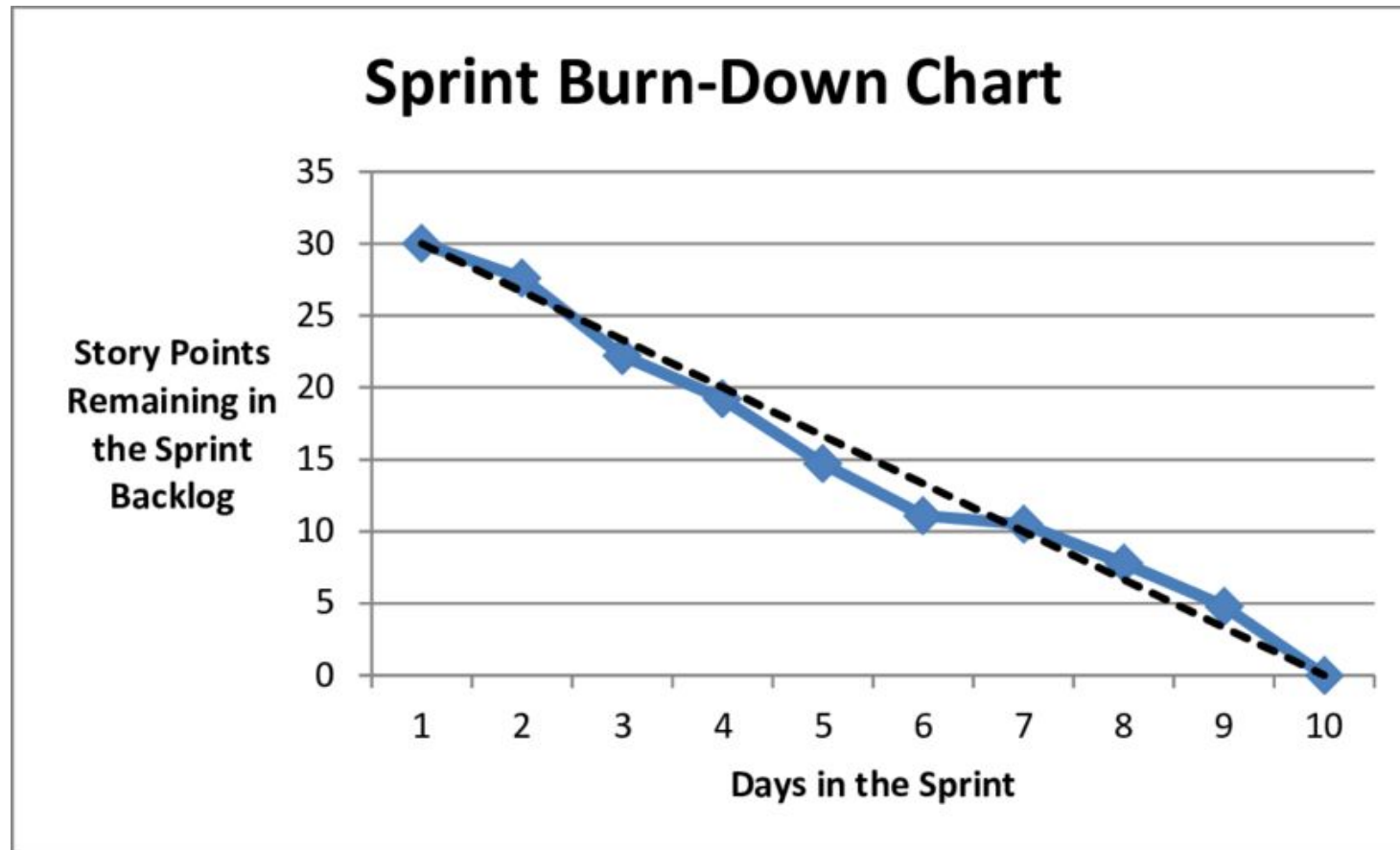
Estimated
velocity = 26

End of sprint



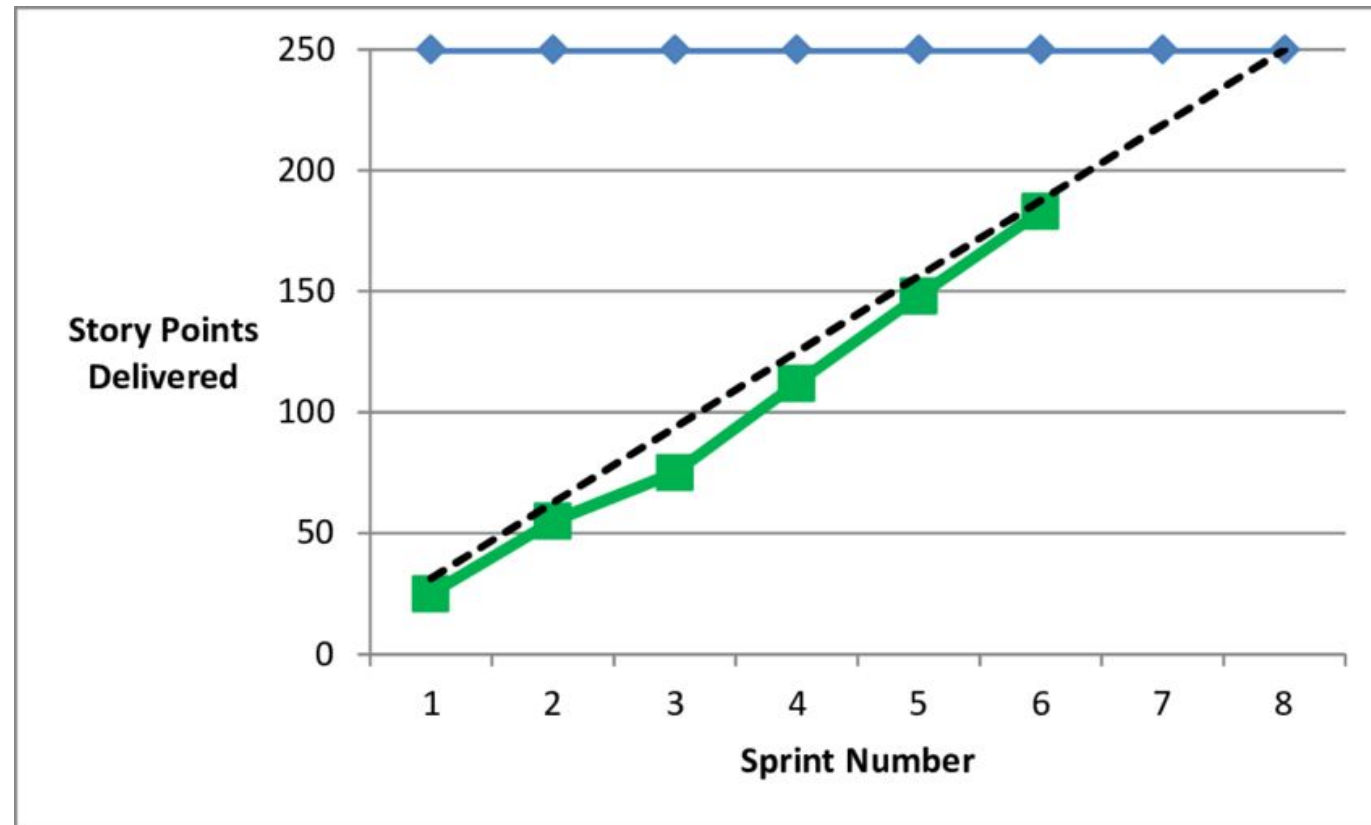
Agile terms.

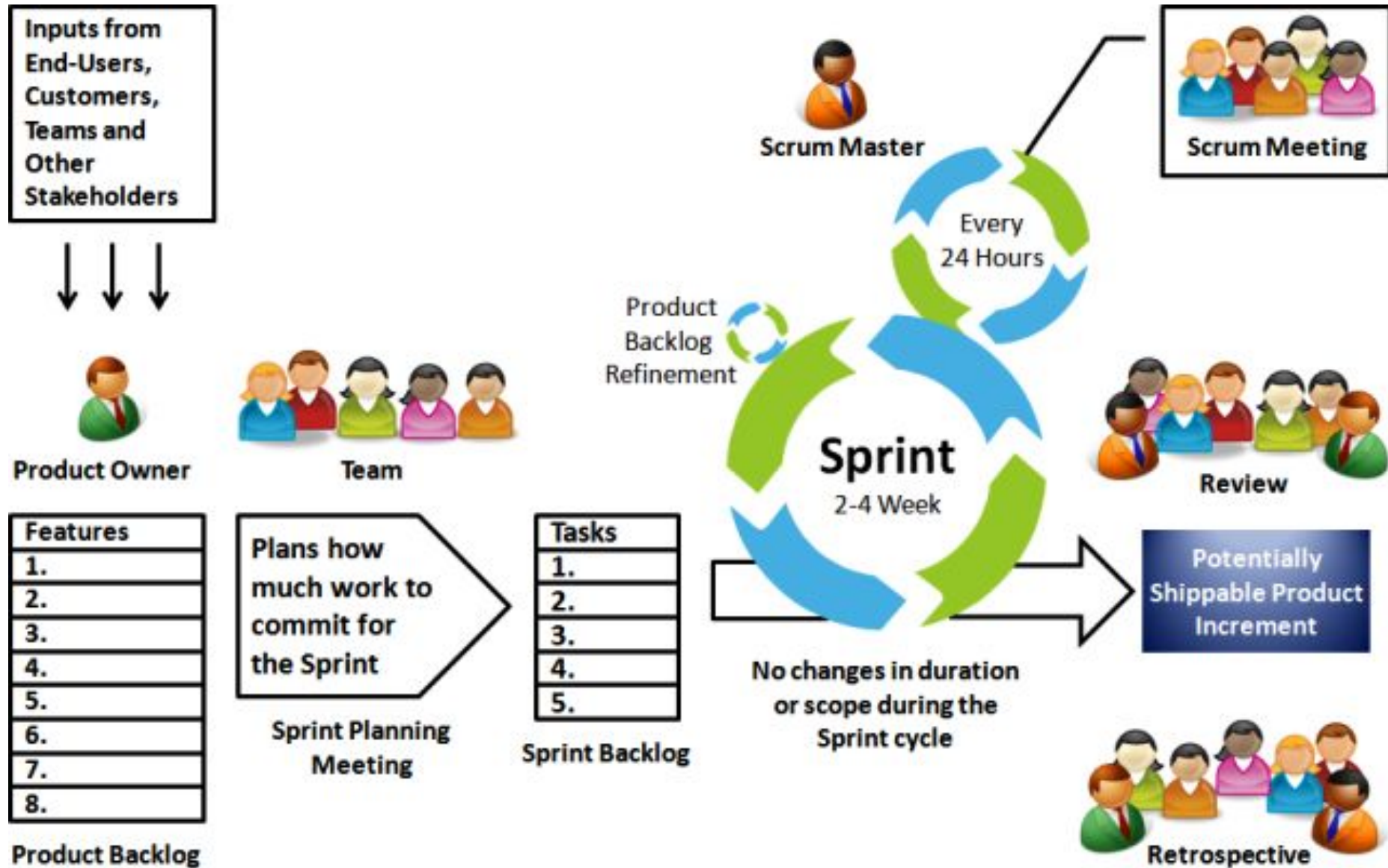
- **Burndown chart:** A burndown chart is an Agile tool, a graphic representation of how fast the team works through the client's user stories.



Agile terms.

- **Burn-up chart:** A burnup chart tracks how much work has been completed. There are two lines on the chart—one line represents total work and the other represents work completed.





Agile team roles.

Product Owner

- Responsible for defining and prioritizing the product backlog.
- Acts as the voice of the customer, ensuring that the team is building the right product by focusing on delivering maximum value.
- Manages stakeholder expectations and sets clear goals for the team.

Agile team roles.

Scrum Master (or Agile Coach)

- Facilitates the Agile process, helping the team follow Agile principles and removing any impediments to progress.
- Coaches the team on Agile best practices and fosters collaboration and continuous improvement.
- Ensures the team remains focused and productive while protecting them from distractions.

Agile team roles.

Development Team

- A cross-functional group responsible for building and delivering the product increment.
- Includes developers, designers, testers, and other specialists needed to complete the project.
- Self-organizing and collectively accountable for delivering high-quality work within each sprint.

Agile team roles.

Stakeholders

- Not part of the core Agile team but play an important role in providing feedback and ensuring the product aligns with business goals.
- Includes customers, business owners, or any external parties with an interest in the project outcome.

Scrum Ceremonies - Meetings



1. Sprint Planning Meeting.

The Sprint Backlog:

is a forecast containing a set of Product Backlog items selected for the Sprint,

a plan for delivering the product Increment and realizing the Sprint Goal,

and contains at least one high priority process improvement (from the retrospective).

Daily Scrum Meeting.

- **Daily Scrum/Daily standup meetings** are a forum for the development team to discuss progress and priorities and work through any issues.
- **Benefits of Daily Scrum Meetings:**
 - o They are used to synchronize team progress by ensuring clarity and visibility of individual performance.
 - o They promote teamwork by sharing not only the progress but also helping each other to overcome challenges.
- **Common mistakes in daily scrum meetings include:**
 - o Tardiness – poor time keeping (start and finish time).
 - o Conversation monopoly – few members talking so much.
 - o Lack of transparency – people not sharing blockers.

3 important questions

What work did you complete yesterday?

1

What have you planned for today?

2

Are you facing any problems or issues?

3



Sprint Review Meeting.

- Occurs at the end of the sprint.
- An opportunity to inspect the increment and adapt the product backlog.
- Demonstrate what is “Done” and receives feedback.
- Team plus key stakeholders.
- Discuss any needed changes to the product backlog.



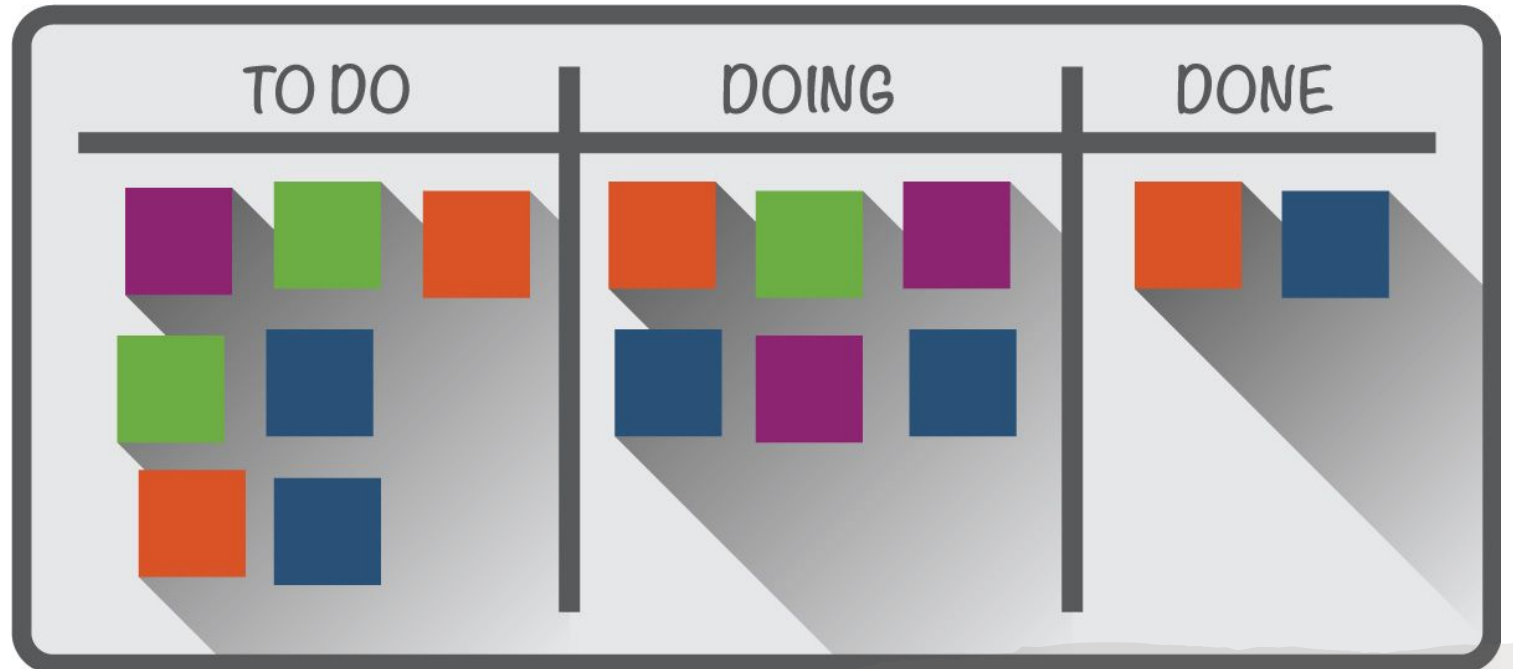
Sprint Retrospective.

- Held at the end of the sprint; After the sprint review and before the sprint planning.
- Purpose:
 - Inspect how the sprint went with regards to people, relationships, processes and tools.
 - Identify and order the major items that went well and potential improvements.
 - Create a plan for implementing these improvements in the next sprint.

AGILE METHODOLOGIES.

- **KANBAN:** A Kanban board (*in Japanese, translates to “Billboard”*) is a simple, visual means of managing projects that enables teams to see the progress so far and what is coming up next. Kanban has two primary purposes which are – visualizing the work of the development team (the features and user stories) and limiting WIP being developed. It segments tasks into three columns:

- **“To Do,”**
- **“Doing,”**
- **“Done.”**



Transparency is key in a dynamic environment. Project tasks should be visible to all - Use Kanban.





ToDo	In Progress	Testing	Done
Installer Update Upgrade Production Server Improve Reliability Preventive Maintenance Optimize Loading Time New Subscriptions Upgrade Automatic Build System Export Item Shop List Features Predictive Analytics Improve User Registration Integrate New Authentication System Refactor UI	New UI Icons Server Load Balancing Revenue Dashboard Add Ads to App Detailed Server Logs New Login Dialog Switch to new Database	Renew Digital Signature Migrate Graphic Assets to Cloud AI Image Recognition Create Alpha 2 Branch Fix Automatic Build Add In App Purchases Test New Performance Profile UI Redesign	Build System Configuration AI Lead Coder Job Desc Item Shop Optimization AI Keyword Sorting Merge Alpha 1 and Dev Branches Dashboard KPI Dynamic Pricing

Conclusion.

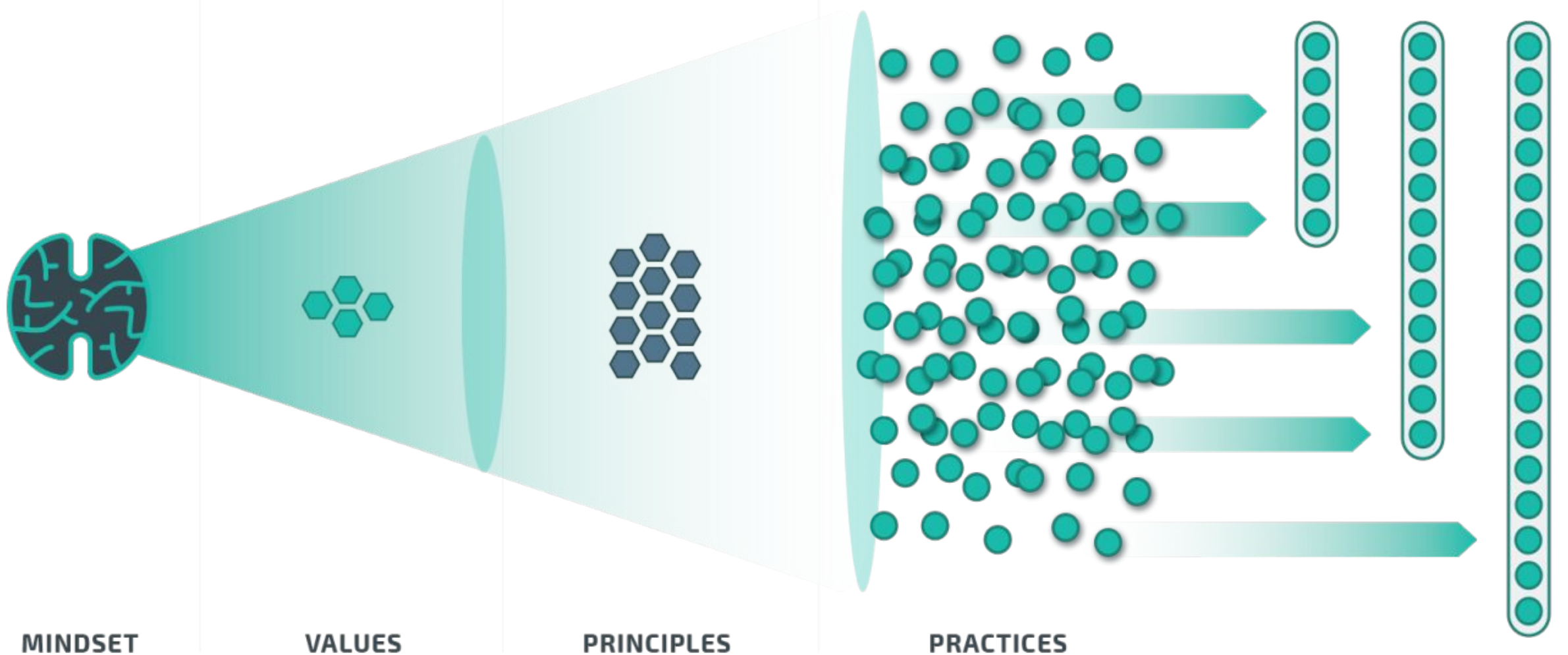
- ❑ Adopting an **adaptive project management approach** is essential in today's dynamic environment due to its ability to handle **uncertainty, enhance customer satisfaction, improve risk management, increase efficiency, boost team morale, and facilitate innovation.**

AGILE IS A
MINDSET

DESCRIBED BY
4 VALUES

DEFINED BY
12 PRINCIPLES

MANIFESTED AS AN
UNLIMITED NUMBER OF PRACTICES



YOUR TAKEAWAY:

An Agile mindset – *a way of thinking that embraces flexibility, collaboration, learning, and adaptability in project management and problem-solving.*